

CSE422 Lab Project Guideline

1. Introduction:

- Select one feature with discrete values as your target feature. [Note: You may discretize one feature with continuous values as well.]
- Give a small introduction to the project and its aim. Describe the problem the project is aiming to solve and the motivation behind the project.

2. Dataset Description:

- Dataset description should include the following:
 - Number of features
 - Number of data points
 - Types of features in the dataset

3. Data Analysis:

- Perform exploratory data analysis to extract some important relationships from the data.
 - [Reference:  EDA Lab CSE422]

4. Dataset Pre-processing:

- Handling null and duplicate values
- Encoding categorical values
- Feature Selection
- Feature scaling
- Other feature engineering, if necessary

[Note: Discuss one problem, and then write about the solutions or pre-processing techniques you have applied to solve that problem.]

5. Model Training:

- You have to apply a Neural Network and at least 2 other models such as KNN, Decision Tree, Logistic Regression, Gaussian Discriminant Analysis, Naive Bayes etc.
- Treat the problem as an unsupervised learning problem and apply k-means clustering.

6. Model Evaluation:

- Compare the models using the following analysis:
 - Confusion Matrix
 - Accuracy, Precision, Recall, F1 Score

- AUC score, ROC curve

7. Conclusion:

- Best model based on the model evaluation
 - Probable reasons behind getting such results

Report Template (Preferred):

- Overleaf:
<https://www.overleaf.com/latex/templates/labreport-ieee-journal-paper-template/ypbfknkvsyhw>